

آزمون: بیوشیمی 4 تیرماه مدت آزمون: 32 بر اساس دفترچه کاربر

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بیوشیمی پایه

۱- کدامیک از کمپلکس‌های زنجیره انتقال الکترون در میتوکندری، مولکول های NADH حاصل از واکنش‌های کاتابولیسمی را اکسید می کند؟

✓ (۱) I

(۲) II

(۳) III

(۴) IV

منبع:

بیوشیمی پزشکی- ج ۲- هیأت مؤلفان- ص ۱۹

۲- در طرفین غشای داخلی میتوکندری، Proton motive force (یا پتانسیل الکتروشیمیایی) در اثر کدام یک از کمیت‌های زیر ایجاد می‌شود؟

(۱) ΔV

(۲) ΔpH

✓ (۳) $\Delta +V\Delta pH$

(۴) $\Delta -V\Delta pH$

منبع:

بیوشیمی پزشکی- ج ۲- هیأت مؤلفان- ص ۲۳ و مطالب کلاس

۳- همه موارد زیر باعث مهار کمپلکس IV زنجیره انتقال الکترون می‌شوند، بجز:

(۱) آزید (N_3^-)

✓ (۲) CO_2

(۳) CO

(۴) CN^-

منبع:

بیوشیمی پزشکی ج ۲- هیأت مؤلفان- ص ۲۲

۴- از اثر کدام دی‌ساکاریداز در روده، دو مونوساکارید گلوکز و فروکتوز تولید می‌شود؟

(۱) مالتاز

[link](#)

1. Which of the electron transport chain complexes in mitochondria oxidizes the NADH molecules produced from catabolic reactions?

1. I ✓
2. II
3. III
4. IV

Source: Medical Biochemistry, Vol. 2, Board of Authors, p. 19

2. On the two sides of the inner mitochondrial membrane, the Proton motive force (or electrochemical potential) is created by which of the following quantities?

1. ΔV
2. ΔpH
3. $\Delta V + \Delta pH$ ✓
4. $\Delta V - \Delta pH$

Source: Medical Biochemistry, Vol. 2, p. 23 and class material

3. All of the following inhibit complex IV of the electron transport chain, EXCEPT:

1. Azide (N_3^-)
2. CO_2 ✓
3. CO
4. CN^-

Source: Medical Biochemistry, Vol. 2, p. 22

4. The action of which disaccharidase in the intestine produces the two monosaccharides glucose and fructose?

1. Maltase
2. Lactase
3. Dextrinase
4. Sucrase (invertase) ✓

Source: Medical Biochemistry, Vol. 2, p. 36

5. Under aerobic conditions, in cells that have mitochondria, the final product of glycolysis is:

1. Lactate
2. Pyruvate ✓
3. Acetyl-CoA
4. CO₂, H₂O, and energy

Source: Medical Biochemistry, Vol. 2, p. 42

6. All of the following are correct, EXCEPT:

1. The final product of glycolysis in red blood cells is acetyl-CoA. ✓
2. The most important regulatory enzyme of glycolysis is phosphofructokinase.
3. Oxygen inhibits glycolysis.
4. AMP stimulates the glycolytic pathway.

Source: Medical Biochemistry, Vol. 2, p. 38

7. What is the main product of the Rapoport shunt in RBCs?

1. 2,3-DPG ✓
2. 1,3-PG
3. ATP
4. PEP

Source: Medical Biochemistry, Vol. 2, p. 41

8. All of the following compounds are inhibitors and/or uncouplers of glycolysis, EXCEPT:

1. Arsenite ✓
2. Arsenate
3. Fluoride
4. 2-Deoxyglucose

Source: Medical Biochemistry, Vol. 2, p. 41

9. Glucose transport into the cells of which tissues are dependent on insulin?

1. Adipose – Liver
2. Liver – Brain
3. Brain – Muscle
4. Muscle – Adipose ✓

Source: Medical Biochemistry, Vol. 2, pp. 48 and 49

10. In which reaction of the Krebs cycle is GTP produced?

1. Oxaloacetate → Malate
2. Succinate → Succinyl-CoA ✓
3. Succinyl-CoA → α-ketoglutarate
4. Isocitrate → cis-aconitate

Source: Medical Biochemistry, Vol. 2, p. 55

11. In which tissue is glycogen broken down only for use by that same tissue?

1. Liver
2. Adipose
3. Muscle ✓
4. Brain and RBC

12. Which glycogen storage disease arises due to deficiency of the debranching enzyme?

1. Von Gierke
2. Pompe
3. McArdle
4. Cori ✓

Source: Medical Biochemistry, Vol. 2, p. 71

13. In which disease is a simultaneous increase in both cholesterol and triglyceride seen?

1. Familial dysbetalipoproteinemia ✓
2. Familial hypercholesterolemia
3. Familial hyperchylomicronemia
4. Apo B100 defect disease

14. Which one plays a role in esterifying free cholesterol in HDL metabolism?

1. LCAT ✓
2. ABCA1
3. ACC
4. CETP

15. Which of the following is seen in liver cells after consuming a carbohydrate-rich diet?

1. Increased ACC activity ✓
2. Decreased FAS activity
3. Increased CPT-1 activity
4. Decreased HMG-CoA reductase activity

16. For the synthesis of cholesterol, all of the following are required, EXCEPT:

1. ATP
2. NADPH+H⁺
3. Malonyl-CoA ✓
4. Acetyl-CoA

17. All of the following compounds participate in the VLDL metabolism pathway, EXCEPT:

1. LDL
2. Chyl (chylomicron) ✓
3. IDL
4. Nascent VLDL

18. Which lipoprotein has the highest percentage of triglyceride?

1. HDL
2. LDL
3. VLDL
4. Chyl (chylomicron) ✓

19. Beta-oxidation of which of the following fatty acids releases the least energy?

1. Linolenic acid ✓
2. Stearic acid
3. Oleic acid
4. Linoleic acid

20. From the beta-oxidation of palmitic acid in the mitochondria, all of the following compounds are produced, EXCEPT:

1. NADPH+H⁺ ✓
2. Acetyl-CoA
3. FADH₂
4. NADH

21. Which one causes inhibition of the enzyme acetyl-CoA carboxylase?

1. cAMP ✓
2. Citrate
3. Insulin
4. Phosphatase

22. Hartnup disease arises due to a defect in the transport of which of the following amino acids?

1. Tryptophan ✓
2. Lysine
3. Arginine
4. Cystine

23. All of the following are synthesized from the amino acid tyrosine, EXCEPT:

1. Dopamine
2. Adrenaline (epinephrine)
3. Melanin
4. Serotonin ✓

24. Which of the following is the coenzyme of transaminases (aminotransferases)?

1. Biotin
2. Thiamine pyrophosphate
3. Niacin
4. Pyridoxal phosphate ✓

25. All of the following amino acids are converted to succinyl-CoA, EXCEPT:

1. Isoleucine
2. Methionine
3. Proline ✓
4. Valine

26. Which of the following compounds is the activator of the regulatory enzyme of the urea cycle?

1. Carbamoyl phosphate
2. N-acetylglutamate ✓
3. Citrulline
4. Glutamine

27. In which of the following conditions is urinary urobilinogen increased?

1. Dubin–Johnson syndrome
2. Biliary duct obstruction
3. Hepatitic jaundice
4. Hemolytic anemia ✓

28. Hereditary deficiency or absence of which enzyme causes immunodeficiency in newborns?

1. Thymidylate synthase
2. Adenosine deaminase ✓
3. Xanthine oxidase
4. Ribonucleotide reductase

29. All of the following compounds participate in the biosynthesis of pyrimidine nucleotides, EXCEPT:

1. Aspartate
2. Phosphoribosyl pyrophosphate (PRPP)
3. Carbamoyl phosphate
4. Glycine ✓

30. 5-Fluorouracil is an inhibitor of which of the following enzymes?

1. Ribonucleotide reductase
2. Dihydrofolate reductase
3. Thymidylate synthase ✓
4. Xanthine oxidase

31. In Hashimoto's disease, how do serum thyroxine (T4) and TSH levels change, respectively?

1. Increase – Decrease
2. Decrease – Increase ✓
3. Increase – Increase
4. Decrease – Decrease

32. In a patient with primary hyperthyroidism, how do blood glucose level and LDL-C level change relative to normal, respectively?

1. Increase – Decrease ✓
2. Decrease – Decrease
3. Increase – Increase
4. Decrease – Increase

33. A patient has a glucagon-secreting tumor. Which of the following metabolic pathways is inhibited under these conditions?

1. Ketogenesis – Protein synthesis
2. Lipolysis – Protein breakdown
3. Gluconeogenesis – Lipolysis
4. Glycogenesis – Lipogenesis ✓